8 On the planet Mars, acceleration due to gravity is $3.71\,\mathrm{m\,s^{-2}}$.

(a) Calculate the time it would take an object to fall to the planet's surface from rest from a point 2 m above the surface. There are no resistances to the motion of the object. **[2]**

(b) The mass of the object is 2.5 kg.

Calculate the magnitude of the normal contact force when the object is at rest on the planet's surface. **[1]**